

Hector Jimenez

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SUMMARY

Electrical Engineer specializing in control systems and circuit design. Experienced in PLC programming, HMI development, and bench testing process control systems. Skilled in power and analog circuit modeling with LTSpice, PCB design with KiCAD, and integrating electrical systems for vehicle and automation applications. Proficient in calibrating RF equipment, troubleshooting PCB faults, and ensuring compliance with industry standards. Hands-on expertise with ABB 800xA and Control Builder M. Experienced in embedded systems programming in C and web app programming in Python and Javascript.

EXPERIENCE

Freelance Hardware Engineer (12/2024 – Present) – Remote

- Designed and prototyped MCU-based hardware using STM32, ESP32, and ATmega platforms.
- Developed schematics and multi-layer PCBs in KiCad for power and embedded systems.
- Built buck/boost converters, battery management, and RS-485/Modbus-RTU interface boards.
- Programmed firmware in C/C++ for peripheral control and data acquisition.
- Performed board bring-up, testing, and validation using oscilloscopes and logic analyzers.

ABB (10/2024 – Present) – Fountain Valley, CA

Hardware Controls Technician

- Wired and commissioned ABB AC800M PLC hardware, including SB822 battery module, CI871A network interface, CI845, TC810, and I/O modules (DI828, DO828, AI825, AO820).
- Configured system architecture and hardware trees in ABB Control Builder M, managing network connectivity and controller applications.
- Designed and deployed custom libraries and function blocks to support automation applications on AC800M controllers.
- Developed and tested Electrical and Process HMI graphics using ABB 800xA and CRISP software.
- Bench-tested ladder logic and HMI integration for Schneider Electric Modicon Quantum PLCs using ProWORX 32, Unity Pro XL, and Wonderware (AVEVA) System Platform software.
- Implemented SCADA systems for a municipal water treatment plant, integrating mixers, pumps, valves, pressure meters, level sensors, flow meters, switchgear, breakers, power monitors, transformers, and ATS

TestCal Instruments (07/2024 – 06/2025) – Diamond Bar, CA

RF Hardware Testing Technician

- Specialized in troubleshooting, maintaining, calibrating, and testing HP 85xx spectrum analyzers, Rohde & Schwarz signal generators, power meters, optical analyzers, and Bluetooth meters using BenchLink.
- Diagnosed and repaired PCB-level faults in spectrum analyzers to restore full functionality.
- Ensured accuracy and reliability of instrumentation and data acquisition systems across medical, aerospace, and telecommunications industries.
- Performed precise RF equipment calibrations for high-frequency applications, maintaining compliance with industry standards.

Nexalure Technologies (12/2023 – 03/2024) – Riverside, CA

Software Engineer Intern

- Built web applications with React front-ends and Python back-ends.
- Designed and optimized RESTful APIs to support scalable applications.

IEEE Solar Car (06/2023 – 03/2024) – Riverside, CA

Electrical Team Lead

- Designed and managed electrical systems for vehicle integration, coordinating tasks across battery, solar arrays, and sensors.
- Conducted component trade-offs for communication technologies and battery management systems.

- Worked with electrical, firmware, and mechanical teams to meet project deadlines.
- Organized workshops on pcb design, layout, and MCU programming to ensure team progress.

Captek Pharma (10/2020 – 09/2021) – La Mirada, CA

Controls Engineer Associate

- Tested process control systems and integrated Cognex Vision for automated X-ray inspection line.
- Supported HMI troubleshooting and process equipment implementations, reducing machine downtime.

Formula SAE Electric (08/2018 – 01/2019) – Riverside, CA

Electrical Systems Team Member

- Designed and implemented the vehicle's electrical architecture, including a custom 20kW three-phase tractive inverter.
- Utilized LTSpice for power and analog circuit modeling and Altium for PCB design.
- Guided team members and maintained key relationships with vendors and sponsors.
- Developed and deployed field-oriented control algorithms in C++ on an embedded STM32 ARM Cortex-M7.

Naval Surface Warfare Center (NAVSEA) (06/2018 – 08/2018) – Norco, CA

Engineering Intern

- Led the complete revamp of a legacy server monitoring application, overseeing design, development, and deployment.
- Developed a C++ application to process and visualize server uptime data in SQL, using the C++ Standard Library and Gnuplot for real-time insights.
- Built Python scripts for data parsing and reporting, alongside Bash automation for server data processing via SSH.
- Collaborated with a cross-functional team, integrating backend and frontend components with the LAMP stack to deliver a scalable, maintainable replacement system.

EDUCATION

University of California, Riverside – B.S. in Electrical Engineering, Power Systems (June 2024)

- EE Coursework: Embedded Systems, FPGA Project Lab, Power Electronics Lab, Signal Processing, Electromagnetics, Circuits and Electronics, Communications, Senior Design (Autonomous Robot), Automatic Control

SKILLS

Software: Altium, KiCad, LTSpice, Solidworks, SQL, ABB 800xA

Languages: C/C++, Python, MATLAB, LabView, Verilog/SystemVerilog